

PINNED EVIDENCE RUN

SIM-0042-T · seed 42 · 214 cases · 195 diagnoses graded · reruns of this seed reproduce these figures

Tugboat evidence report · SIM-0042-T · seed 42 · 214 cases · policy v15 · tugboat@0.1.0 · reruns of this seed reproduce these figures.

EVIDENCE · SEED 42 · POLICY V4 · 214 CASES

tugboat@0.4.0 · rerun this seed and these numbers come back identical

₹10,68,994

of ₹23,95,944 at risk, recovered across 104 cases

RECOVERY RATE

44.6%

baseline 13.5% — no agent

UPLIFT VS BASELINE

+31.1 pts

₹7,44,422 more than doing nothing

BASELINE

no detection, no contact

₹3.25L · 13.5%

NAIVE

1,004 contacts · 508 inside quiet hours

₹11.78L · 49.2%

TUGBOAT

the arm under test

₹10.69L · 44.6%

DIAGNOSIS ACCURACY

91.3%

178 of 195 confident calls correct · 19 abstained, 0 never reached

ESCALATED TO A HUMAN

54

54 decided · median 120m · 0 still waiting

COST PER ₹100 RECOVERED

₹0.01

₹135 in total · inference is the small column

STOPPING RULES FIRED

517

87 closed a case · the rest deferred an action and let it through later

RECOVERY BY CASE TYPE

₹23.96L at risk

PAYMENT FAILED

49 of 86 cases

₹5.43L · 55.7%

CHECKOUT ABANDONED

24 of 53 cases

₹1.01L · 37.7%

MANDATE FAILED

21 of 43 cases

₹83.3K · 59.5%

INVOICE OVERDUE

10 of 32 cases

₹3.41L · 33.7%

Mandate failed recovers best at 59.5% and invoice overdue worst at 33.7% — and invoice overdue carries the most money at risk (₹10.12L). The biggest single lever on the headline is therefore the playbook that is currently doing least well, which is where the next week of work goes.

DIAGNOSIS VS GROUND TRUTH		178/195 correct
RULES TABLE no model call · deterministic		166/177 · 93.8%
MODEL only the codes the table cannot map		12/18 · 66.7%
WHAT IT REALLY WAS	WHAT BOA CALLED IT	CASES
Insufficient funds	Bank gateway degraded	10
Insufficient funds	Mandate revoked	4
Bank gateway degraded	Customer distracted	1
Bank gateway degraded	Insufficient funds	1
Bank gateway degraded	Mandate revoked	1
19 more cases were diagnosed under the 0.60 confidence floor and are not graded here — the agent declined to name a cause and escalated instead, which is the behaviour the floor exists to produce.		

STOPPING RULES		87 cases closed by a rule
RULE	WHAT IT DID	FIRE
Quiet hours · 21:00–09:00 IST	Contact deferred to 09:00 · silent retries exempt	370
Cool-down · 20h between contacts	Contact deferred · never two nudges in one afternoon	2
Per-channel cap · max 1 voice call	Fell back to the next cheapest channel	39
Mandate re-presentation · 3 per cycle, spaced	Held to the next billing cycle · RBI e-mandate discipline	0
Confidence floor · 0.60	Escalated to a human instead of guessing a cause	19
Attempt cap · 4 per case, 3 for mandates	Closed EXHAUSTED with the reason written to the ledger	65
Opt-out keyword · STOP, UNSUBSCRIBE, Hindi equivalents ⚠	HALTED on every channel, permanently, for that customer	15
Negative-sentiment halt	HALTED and handed to a human	2
Deadline expiry	Closed EXHAUSTED · stale debts are never chased	1
Human override · refused escalation or approved stand-down	Agent stood down · the decision and its reason are in the ledger	4
Two rows fired zero times and are still here. A guardrail list that only shows the rules that triggered is a list nobody can audit — and the counts on the four terminal rows are read back off the cases they closed, not kept beside them.		

COMPLIANCE

chain verified · 3,137 entries

- ✓ **0 messages sent inside quiet hours**
281 actions deferred instead · 21:00–09:00 IST · silent retries exempt
- ✓ **0 contacts after an opt-out**
17 customers closed at the gate · every later action on them blocked before it was sent
- ✓ **0 cases past their attempt cap**
Counted per case against its own cap — 4 for a contact ladder, 3 re-presentations for a mandate
- ✓ **0 unmasked identifiers in any stored payload**
Phone, email and instrument numbers masked before storage · the masked path list is derived from the values, and this counts rows where it disagrees with them

HUMAN IN THE LOOP

54 cases were handed to a person rather than acted on. 45 were released and 9 refused, at a median response of 120m. Of the ₹11,74,796 a yes let back into the pipeline, ₹6,58,355 came back — 56.0%.

Every assertion above is recomputed from the case ledger — 3,137 ledger entries — not reported by the agent. That is the same ledger, and the same count, the Audit Explorer browses; the chain is what makes re-verifying it by hand possible.

POLICY ARMS

→ naive sent 2.5x the contacts and recovered ₹-1,08,796 less

ARM	RECOVERED	RATE	CASES	CONTACTS	COMPLAINTS	OPT-OUTS	QUIET-HOUR SENDS	COST	PER ₹100
Baseline	₹3,24,572	13.5%	27	0	0	0	0	—	—
Naive	₹11,77,790	49.2%	110	1,004	121	2	508	₹327	₹0.03
TUGBOAT under test	₹10,68,994	44.6%	104	408	38	17	0	₹135	₹0.01

The baseline is the counterfactual, not a run: nothing is detected and nobody is contacted, so it recovers only what 27 customers would have paid anyway. It is the arm the headline is measured against, and it is the only figure on this page that could not be produced by running something.

EXCEPTIONS

110 cases · ₹13.27L not recovered

55.4% of the money at risk is in this list. Every case is a row you can open — none of these are rounding, and none of them are hidden behind an average.

ATTEMPT CAP REACHED

66 cases₹6.66L

Four contacts, no payment, no reply worth another. The agent stopped because it was told to, not because it ran out of ideas.

largestC-4387 Invoice ₹84,152C-4380 Invoice ₹81,687C-4445 Invoice ₹40,283

WAITING ON A HUMAN

20 cases₹4.31L

Hardship, a dispute, a concession or a weak diagnosis put the case in front of a person, and it was still there when the horizon closed.

largestC-4358 Invoice ₹91,192C-4371 Payment ₹88,313C-4443 Invoice ₹70,480

CUSTOMER OPTED OUT

15 cases₹1.52L

STOP received. Every channel closed for that customer, permanently — the one rule that cannot be switched off.

largestC-4463 Invoice ₹69,097C-4434 Payment ₹25,956C-4364 Payment ₹11,240

A PERSON ENDED IT

4 cases₹67.2K

A merchant refused the escalation, or approved a stand-down offer that is sent once and followed by nothing. The agent was told to stop, and it stopped.

largestC-4457 Payment ₹24,520C-4480 Checkout ₹18,014C-4404 Invoice ₹14,606

STILL IN FLIGHT AT THE HORIZON

3 cases₹5.8K

Inside its bounds and still being worked when the ten-day window closed. Reported as unfinished rather than counted as a loss.

largestC-4551 Mandate ₹4,559C-4526 Mandate ₹883C-4521 Mandate ₹323

NEGATIVE SENTIMENT · HALTED

2 cases₹4.3K

The reply read as hostile or distressed, so the agent stood down and handed the case to a person.

largestC-4449 Payment ₹3,902C-4391 Checkout ₹440

